

Command Line Arguments in Java

Command Line Arguments are arguments passed to a Java program at runtime using the command line. These arguments are stored in the `args` array of the `main()` method and can be accessed by the program during execution.

Structure of the `main()` Method

The `main()` method in Java is defined as:

```
public static void main(String[] args)
```

- `args`: An array of `String` objects that stores the command line arguments.
 - The arguments can be accessed using indices: `args[0]`, `args[1]`, etc.
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How to Pass Command Line Arguments

1. **Write a Java Program:** Save your program in a file, e.g., `CommandLineExample.java`.
2. **Compile the Program:** Use the `javac` command to compile:

```
javac CommandLineExample.java
```

3. **Run the Program with Arguments:** Use the `java` command to execute and pass arguments:

```
java CommandLineExample arg1 arg2 arg3
```

Here:

- `arg1`, `arg2`, `arg3` are the command line arguments.
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Example 1: Print Command Line Arguments

```
public class CommandLineExample {  
    public static void main(String[] args) {  
        System.out.println("Number of arguments: " + args.length);  
    }  
}
```

```
        for (int i = 0; i < args.length; i++) {
            System.out.println("Argument " + i + ": " + args[i]);
        }
    }
}
```

Steps to Run:

1. Compile:

```
javac CommandLineExample.java
```

2. Execute with arguments:

```
java CommandLineExample Hello World 2024
```

Output:

```
Number of arguments: 3
Argument 0: Hello
Argument 1: World
Argument 2: 2024
```

Example 2: Adding Two Numbers Passed as Arguments

```
public class AddNumbers {
    public static void main(String[] args) {
        if (args.length < 2) {
            System.out.println("Please provide two numbers as arguments.");
            return;
        }

        // Parse arguments to integers
        int num1 = Integer.parseInt(args[0]);
        int num2 = Integer.parseInt(args[1]);

        int sum = num1 + num2;
        System.out.println("The sum is: " + sum);
    }
}
```

Steps to Run:

1. Compile:

```
javac AddNumbers.java
```

2. Execute with arguments:

```
java AddNumbers 5 10
```

Output:

```
The sum is: 15
```

Key Points to Remember

- Command line arguments are always passed as strings.
- Use `Integer.parseInt()` or `Double.parseDouble()` to convert arguments to numeric types.
- Always validate input to prevent runtime errors.